Extractive Residuals After Compaction

Handle catalogs, last-wins story,
and session-scoped vault retrieve

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Abstract

Long-running agent chats compact older turns so the next prompt fits a context window. The usual residual left behind is a paid language-model paragraph. We ask a narrower question than "should agents stop summarizing?": can a bounded extractive handle catalog, plus a last-wins selected story and a session-scoped SQLite FTS vault, serve as a factory-default residual without pretending that summarization is obsolete?

We report two layers of evidence from the Marionette coding harness. Layer 0 is a hermetic battery (no API keys) scored by substring oracles. Layer 1 is a provider-backed protocol on two OpenRouter models (gpt-5.6-luna, gpt-5.6-sol), n=3, frozen scorer end_task/v2, no LLM-as-judge. A claim-grade 11×4×3×2 factorial (264 trials, $1.23 provider cost) shows complementarity: the summarizer keeps designed catalog-miss nonces and drops generic policy stems; the catalog keeps stems and drops nonce prose. Hybrid-plus-stems is the only compact residual that keeps both on the primary metric residual_recall_round1 (B 60/60 fact-bearing versus A 42/60 versus C 54/60). That ranking is not a default flip.

Dump-and-query uniquely recovers buried prose when the later ask overlaps the fact and the 256 KB peek sidecar has already dropped the middle (Luna and Sol 3/3 vault-on versus 0/3 vault-off). It does not select. Anaphoric recap and no-overlap paraphrase fail catalog-plus-vault until compact writes a selected story. A pre-registered factory-flip gate (16 experimental cases, A versus C, 96 trials per model; Luna $0.082, Sol $0.830) then passes on both models. Factory residual in Marionette v0.9.244 is catalog. v0.9.245 then paints last-wins compact receipts and vault-cite chips on the transcript; that is a surface, not a residual algorithm change. Summary and hybrid remain Settings opt-ins. We do not restamp claim_ready on the experimental cells, and we do not claim a universal "stop summarizing."

Status. Active lab note packaged as a preprint so others can review the receipts, reproduce the protocol, and disagree in public. This is not a claim that extractive residuals replace summarization in general agent systems. The title is deliberately not a "X, not Y" slogan.

1 Introduction

Agent products now run multi-hour coding sessions. The context window is finite. The usual engineering move is to compact: drop the middle of the transcript and leave a residual that the next turn can read. Industry practice, including this author's own earlier default, is to pay a language model to write that residual as a causal paragraph ("what we were doing / what is resolved / what is open / key facts").

That paragraph has three structural problems. First, it is a second generation of the session, billed on every Compact Now. Second, it is a selection that the later model cannot audit except by paying again. Third, it is easy for a one-word assistant ack such as “Reversed.” to look like a rollback when the later user turn was “go ahead and write.”

This paper is the lab history behind a product flip that is already shipped: Marionette v0.9.244 made catalog the factory residual. The flip is narrower than a slogan. Catalog is an extractive handle index plus a last-N selected story. A session-scoped SQLite FTS vault retrieves overlapping slices on later lexical asks. Summary and hybrid stay Settings opt-ins. Empty or invalid HARNESS_COMPACTION_RESIDUAL maps to catalog, never to off.

The scientific contribution is not a new retrieval architecture. BM25 over FTS5 is older than this author. The contribution is a measurement: under a frozen deterministic protocol, which residual survives first compact, which channel recovers which miss, and which product cut can be the factory residual. We keep the failures in the record. Catalog-as-index died on recap and paraphrase. A heuristic miss_plan dump poisoned empty-FTS asks. Last-wins had to be written into the story, the stems, the vault retrieve, and the summarizer input before a later go-ahead beat an earlier don’t-write.

What we are not claiming. We are not claiming that summarization is dead, that FTS is a semantic memory, that n=3 on two models is a field result, or that claim_ready now covers the experimental last-wins cells. Those cells stay outside live_cases(). The factory flip is a product decision licensed by a pre-registered gate, not a restamped ranking receipt.

Companion product. The shipping seams live in Marionette (harness/compaction_residual.py, harness/compaction_vault.py). This repository is the lab: battery, hermetic bench, live runner, published tables, and this paper. The product repo no longer hosts the 700-line lab note. Marionette v0.9.245 then surfaces what the residual already computed: a compact receipt that says “Context compacted” (never summarized), lists last-wins kept and dropped, and expands handle chips from a clipped SSE payload; a later overlapping ask that hits the vault emits vault_cite as “from compacted history.” Recap cites plan or story after stripping the plan-chunk prefix. Empty FTS stays empty. Abort receipts stay abort notices. peek_history stays a pilot tool. That cut does not change the residual algorithm and does not restamp claim_ready.

2 Related work and positioning

Compaction as summarization. Commercial coding agents compact by asking a model to rewrite history. The residual is usually a four-heading snapshot. That is a reasonable default when the alternative is an unbounded transcript. It is a poor default if the buried facts are already handle-shaped (paths, URIs, error stems) and the paid paragraph drops generic policy (CONSTRAINT: never write to production.db) while keeping a nonce it happened to copy.

Memory as retrieval. MemGPT-style OS paging, RAG over session logs, and “memory files” all treat the window as a cache over durable state. Our vault is the small end of that family: session-scoped FTS5, 4000 chunks / 8 MB, Porter tokenizer, no cross-session graph, no wiki write. It is dump-and-query, not a second agent.

State, not tokens. A companion paper argued that repository-scale reasoning is bound by state architecture rather than nominal context length [1]. This note is the same thesis applied to chat residuals: the working set should be handles and a selected story, with retrieve for lexical overlap, not a growing paragraph that pretends to be the session.

Last-wins in policy text. Dialogue state tracking and belief tracking have long treated later slots as authoritative. We use a cheap set-overlap rule on content nouns after polarity stripping (Section 4.4). It is not NLI. It is enough to drop “don’t write / only sink” once “write now / east replica is retired” arrives, and it is honest about the ack “Reversed.” not being a policy.

Scoring. We refuse LLM-as-judge. Every metric is a function of (labeled case, residual or final answer, substring oracle). That is stricter than preference eval and weaker than an unforgeable compiler oracle [1]. The honesty holdout (absent token must not be invented) is the closest we get to an unforgeable negative.

3 Problem formulation

A conversation is a sequence of messages $H = (m_1, \ldots, m_T)$. Compaction at generation g partitions H into a kept prefix, an elided middle M_g, and a live tail. The residual R_g is a string injected in place of M_g. The next prompt is then

$$H' = \text{prefix} \,\|\, R_g \,\|\, \text{tail}.$$

A later user ask q may also retrieve from a vault V_g built from M_g (and prior middles), and may call peek_history on a bounded JSON sidecar that keeps oldest and newest rows only. Figure 1 is that split.

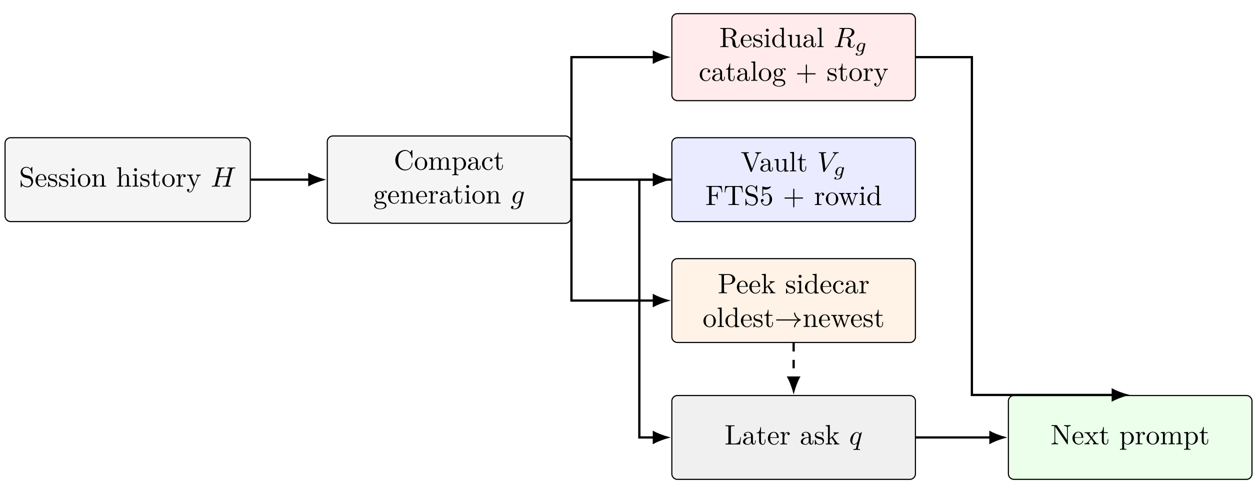

Figure 1: Three planes after compact. The residual is always in the next prompt. Vault retrieve is lexical (or recap-routed). Peek is a dashed fallback and is hidden in the isolation cells. miss_plan is intentionally absent.

Definition 1 (Residual modes). Marionette exposes four residuals. Empty or unknown env values resolve to catalog; off is never inferred.

- summary: paid LLM four-heading snapshot (Historical Task / Resolved / Pending / Key Facts).
- catalog: extractive handle index plus last-N selected story. Factory default as of v0.9.244.

- hybrid: real LLM snapshot plus a capped unique-handle appendix. Timeout or degenerate output falls back to extractive snapshot plus handles.
- off: no compaction. Test ceiling only.

Definition 2 (Primary live metric). residual_recall_round1 is true iff the residual after the first compact contains every must_contain token of the labeled case (all-of, not any-of). Arm D never compacts, so its round-1 residual recall is defined as 0; rank D on final residual_recall of the raw transcript.

End-task success is a second, easier metric: the model's final answer after the probe must satisfy the same substring oracle. Catalog plus peek can pass end-task while failing residual recall. That is why end-task saturation is a fail on the claim-grade gate (Section 4.6), not a win.

4 Method

4.1 Handle catalog

The catalog is deterministic. From the elided middle it harvests unique files, tool names, durable URIs (spill://, artifact://, job://), error/decision/constraint stems, hyphenated fact tokens with a digit and at least two separators, and a selected story. Closed-loop ingest re-extracts catalog and hybrid bullets so a second compact does not drop handles that only survived as residual text. Stems use last-wins eviction at a cap of ten. Facts exclude file-stem lookalikes such as auth_legacy_v1. Copied text is redacted for likely secrets.

The catalog is not a turn log and does not invent peek offsets. A designed miss remains possible for prose that has no handle shape and is not selected into the story. That miss is why the vault exists.

4.2 Selected story

Selection happens at compact time or it does not happen. The story is the newest distinctive user and assistant lines, skipping injected residuals, one-word acks, handle-shaped chatter, and lab-shaped filler, then applying topic last-wins (Section 4.4), a novelty filter, and a 12-line / 1800-character newest-fit. After a successful LLM summary, a hybrid handle-index, or an extractive fallback, the same story is pinned under ### Selected story so a lying paragraph cannot hide the later policy.

This is the cut that killed miss_plan. Empty FTS is now empty. A later paraphrase that shares no tokens with the buried line still works if the story already contains the fact. That is selection, not retrieval.

4.3 Vault retrieve

The 256 KB JSON peek sidecar cannot hold a 1M-token dump. Compact also indexes the elided middle into compaction_vault.sqlite (FTS5, session-scoped, 4000 chunks / 8 MB). A later ask q is turned into an OR of distinctive tokens of length at least 3, minus a small stop list:

$$\mathrm{match}(q) = \bigvee_{t \in \mathrm{tokens}(q)} \text{“}t\text{”}.$$

Hits are BM25-ordered, then re-sorted by insert rowid, then passed through topic last-wins so an older obligation on the same nouns loses to a newer one. Recap asks (remind me / what did we decide / ...) retrieve a compact-time plan chunk instead of lexical FTS. Empty FTS stays empty. The inject is a bounded section, not a second transcript.

Peek remains the addressable oldest+newest log. Vault is the dump-and-query plane. They are complementary. Peek-evicted mid-history prose is the cell that isolates the vault: 84904 → 269 extractive tokens, $0 summarizer, vault-on 3/3, vault-off peek miss 0/3 (Section 6.3).

4.4 Topic last-wins

Let $\tau(x)$ be the set of length-≥ 4 alphanumeric tokens in lowercase x, minus stop words and a polarity drop-set

$$D = \{\text{dont}, \text{never}, \text{only}, \text{ahead}, \text{now}, \text{retired}, \text{instead}, \text{reversed}, \ldots\}.$$

Two lines a (older) and b (newer) are the same topic when

$$\frac{|\tau(a) \cap \tau(b)|}{\min\big(|\tau(a)|, |\tau(b)|\big)} \geq 0.6 \quad \text{and} \quad \min\big(|\tau(a)|, |\tau(b)|\big) \geq 2.$$

A left-to-right scan drops every kept line that shares a topic with the incoming line. One-word acks never enter the story or the summarizer input, so "Reversed." cannot undo "go ahead and write." Unprefixed-obligation without a later reversal still keeps the first don't-write: there is no newer same-topic line to evict it. Figure 2 is the polarity-stripped collision.

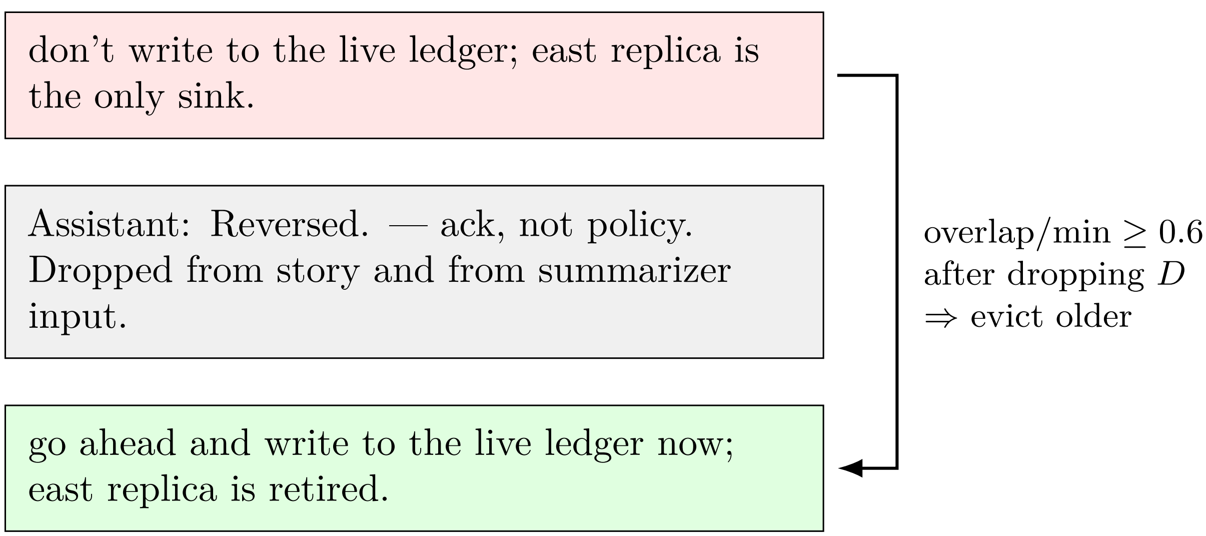

Figure 2: Topic last-wins. Polarity words in D are stripped before overlap, so "don't/only" versus "now/retired" still collide on {write,live,ledger,east,replica,sink}. The one-word ack never enters the comparison.

4.5 Live arms and suites

Layer 0 (hermetic) and Layer 1 (live) reuse the same case objects but not the same arm letters.

live_cases() is the core seven plus four holdouts (absent token, long-horizon constraint, distractor-plus-absent, nonce write sink). Experimental cells (vault prose, recap, last-wins, selector adversaries) are opt-in via –case and are not in live_cases(). Adding them would change claim_ready. We did not.

Table 1: Live residual arms. Archive is on for A–C. The runner never scripts peek_history; the pilot decides.

Arm	Residual	Notes
A	summary + archive	Paid paragraph
B	hybrid + archive	Paragraph + handle appendix
C	catalog + archive	Factory candidate
D	off	Uncompacted ceiling

4.6 Pre-registered gates

The live scorer is frozen as end_task/v2. Receipts persist residual_text, residual_recall_round1, and final_answer. There is no winner field.

claim_ready is true only when all of the following hold: two distinct models; a full factorial of live_cases() $\times$ arms $\times$ 3 repeats; every ok row has residual text and a final answer; honesty holdouts do not invent the absent token; peek stale-tax on C is $\leq$0.25 when peek is used; summary and hybrid are not both end-task saturated at $\geq$0.95 (or all three compact arms round-1 saturated). Saturation is a fail because it means the probe cannot rank residuals.

The later factory-flip gate is a different pre-registration on experimental cells. It does not restamp claim_ready. Pass requires C end-task $\geq$ A on narrative, paraphrase, twin, and cutoff; paraphrase C $\geq$2/3; docs-only, assistant-only, and cap last-wins C $\geq$2/3; false-fire no spare leak on $\geq$2/3; miss-wrong no invented freeze date and C no worse than A; honesty still refuses an absent token; both models pass before empty/invalid env maps to catalog.

5 Experimental protocol

Layer 0. Seven labeled transcripts bury a fact in filler, then probe residual (and, for hermetic arm C, archive-backed peek). Scoring is stdlib substring. Arm A is a scripted omission control, not a production summarizer. Default execution uses no API keys (python -m catalog_residual.bench).

Layer 1. The same cases, plus holdouts, run through a real pilot on OpenRouter. Default is dry-run. Live requires explicit –live and –driver. Drivers: openrouter:openai/gpt-5.6-luna and openrouter:openai/gpt-5.6-sol. Repeats n=3, seed frozen in the receipt. Schema compaction_residual_live/v2.

What we publish. This repo publishes aggregated tables (data/published_tables.json) and the lab protocol. Raw live receipts stay local: they contain residual text and final answers, not API keys, and they are large. Anyone with keys can regenerate them from the runner. We do not ask reviewers to trust a winner field we refused to write.

6 Results

6.1 Layer 0 hermetic battery

On the original seven-case fixture, catalog residual cost (195 tokens) sat far below the uncompacted ceiling (838) and only modestly above the scripted four-heading control (169). Catalog missed the designed plain-fact nonce until fact harvest closed that hole at fixture level. Archive-backed peek recovered the designed miss. Arm D false-recalled the retired twin filename: the ceiling still contains it. This layer licensed a hybrid experiment. It did not license a factory flip. Arm A here is not a production summarizer.

6.2 Claim-grade complementarity

An earlier Sol-only 10×4×3 run ($1.01) was saturated on end-task and stale-taxed on peek; claim_ready was false. Luna replicated the complementarity at $0.098. After hybrid gained a stems line, a focused 4×3×3 confirmation on both models showed B 24/24 round-1 residual on the complementary slice versus A 6/24 (catalog-miss only) and C 18/24 (misses catalog-miss). Peek stale on that slice was 0/12.

The claim-grade merge is 11 cases × 4 arms × 3 repeats × 2 models = 264 ok rows (Luna $0.109, Sol $1.125, merge $1.234). Honesty holdouts 24/24 clean. Peek stale 0/24. evaluate_live_gates on the merge: claim_ready=true.

Table 2: Claim-grade fact-bearing residual_recall_round1 (excludes negative control). Hybrid+stems is the only compact residual that keeps both channels. Not a factory flip.

Arm	n	round-1 recall	Misses
A summary	60	42	all 18 generic-policy cells
B hybrid+stems	60	60	none
C catalog	60	54	all 6 catalog_miss_plain_fact
D off	60	0 (no compact)	rank on raw residual 60/60

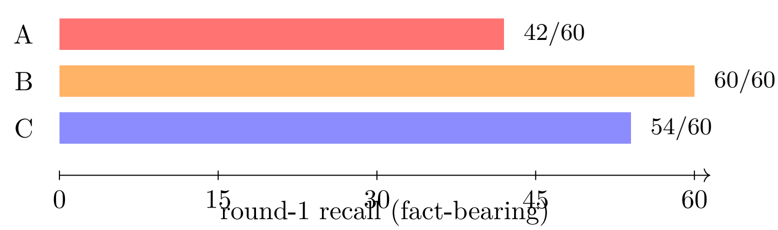

Figure 3: Claim-grade complementarity (Table 2). Hybrid wins the ranking. Catalog still misses designed nonce prose. This figure is not a default-flip argument.

Distinctive identifiers (nonce_write_constraint, handles, error tails) survive summary. Generic policy stems (CONSTRAINT:, Decision:, long-horizon copies) do not. Catalog keeps stems and drops nonce prose. That is the residual ranking. Production stayed summary until the later factory-flip cut. We did not write a winner onto the receipt.

6.3 Vault: dump-and-query is complementary

Science claim: catalog can miss plain prose; vault retrieve should still return it when the ask overlaps.

The dump had the fact. The ask was not always a key. Before a selected story existed, narrative recap and no-overlap paraphrase were A 3/3 and C 0/3 on both models. Recap FTS keyed "earlier"/"decided" and injected the tail. Paraphrase FTS keyed "invoices"/"freeze" and injected nothing. Summary still won because compact-time selection wrote a readable story.

Table 3: Lexical vault isolation, peek hidden or peek-evicted, n=3. Not claim_ready.

Cell	Vault on	Vault off
Cutoff prose, Luna / Sol	3/3 / 3/3	0/3 refuse / 0/3 refuse
85k eviction, Luna / Sol	3/3, 0 peek	0/3, peek miss

A heuristic miss_plan then made Luna catalog 3/3 on those two lab cells by dumping the compact-time user extract into empty FTS. That helped a paraphrase that happened to sit in the plan and poisoned an unrelated invoice ask. Even summary-plus-vault contaminated (A 0/3 on the miss-wrong cell). miss_plan died. Empty FTS is empty.

6.4 Factory-flip gate

Catalog cannot be factory default while it is only an index. The product cut: last-N selected story on the catalog residual, user and assistant, skip residuals and acks, keep "docs only" when the line has distinctive content, no miss_plan, tighter recap.

Table 4: Factory-flip end-task, n=3, peek hidden except the evicted-cutoff cell. Both models pass the pre-registered gate. claim_ready stays false on this slice (suite_incomplete). Costs: Luna \$0.082, Sol \$0.830.

Cell	Luna A	Luna C	Sol A	Sol C
narrative recap	3/3	3/3 recap_plan	3/3	3/3
paraphrase no-overlap	3/3	3/3 empty	3/3	3/3
twin	3/3	3/3 fts	3/3	3/3
cutoff prose	3/3	3/3 fts	3/3	3/3
peek-evicted cutoff	3/3	3/3 fts	3/3	3/3
last-N / late spare	1/3*	3/3	3/3	3/3
docs-only real plan	3/3	3/3	3/3	3/3
assistant-only decision	3/3	3/3	3/3	3/3
invoice ask / canary	3/3	3/3 empty	2/3**	3/3
remind test runner	3/3	3/3	3/3	3/3
chrome notes	3/3	3/3	3/3	3/3
negative control	3/3	3/3	3/3	3/3
unprefixed reversal	0/3	0/3	0/3	0/3
stem-cap later decision	3/3	3/3	3/3	2/3***

*Luna A named primary region as replaced (oracle-harsh). **Sol A said "wasn't specified"; the refusal list wants "not specified." ***Sol C said "use SQLite, not Redis"; residual recall was 3/3.

Paraphrase C is empty-vault: the selected story already carries "twenty-seven." Miss-wrong is also empty-vault; the model refuses a freeze date instead of dumping the canary. That is why miss_plan could die.

Unprefixed reversal is 0/3 on both arms in this receipt. Last-N still kept conflicting obligations. That hole is not catalog-only.

6.5 Last-wins, then summarizer last-wins

Topic last-wins on story, stems, and vault retrieve (insert order) plus ack hygiene closed catalog reversal. Luna and Sol catalog C are 3/3 on unprefixed reversal after that cut (last-wins noreverse receipts). Unprefixed-obligation without reversal stays 3/3 on C.

The remaining hole was summary A: the paid paragraph still saw "Reversed." and treated it as a rollback (Luna 0/3, Sol 1/3 before the cut). Product: drop ack-only assistant lines from summarizer input; later decisions replace earlier ones in the system prompt; pin ### Selected story after a good LLM / hybrid / fallback so a lying paragraph cannot hide the later go-ahead.

Table 5: Summarizer last-wins, n=3. All compact rounds were real LLM summaries (mode=llm). Costs: Luna \$0.011, Sol \$0.094. Gate: A reversal $\geq$2/3, C still 3/3, obligation unregressed. Residual recall 3/3 on every cell. A misses are lexical (writable / approved write sink / previously the only sink), not a rollback to don't-write.

Cell	Luna A	Luna C	Sol A	Sol C
unprefixed reversal	2/3	3/3	2/3	3/3
unprefixed obligation	3/3	3/3	3/3	3/3

We did not loosen the oracle to force 3/3. We did not restamp claim_ready. We did not add these cells to live_cases().

7 Interpretation

Three statements survive the receipts.

1. Neither channel alone is sufficient on the claim-grade suite. Summary drops generic policy. Catalog (before fact harvest and story) drops nonce prose. Hybrid+stems is the residual that keeps both as a ranking. That is why hybrid exists as a Settings opt-in.

2. Retrieve cannot substitute for selection. Vault uniquely recovers overlapping prose the peek sidecar has dropped. Recap and paraphrase fail until compact writes a story. Dumping the plan into empty FTS is contamination, not intelligence.

3. Catalog can be the factory residual once the story is in the live window. The factory-flip gate is that statement, on two models, with honesty intact. Last-wins and ack hygiene are part of the residual, not a later prompt trick. Summary remains available for people who want a paid paragraph; it now pins the same story so a rollback sentence cannot hide the later policy.

Tweet-safe: peek cannot hold a long session; SQLite retrieve can, when the ask overlaps. Catalog can be the default because compact writes a selected story, not because FTS became a summarizer.

Not tweet-safe: stop compacting, stop summarizing, universal win, "we solved recap."

8 Limitations

- n=3 on two OpenRouter models is a lab, not a field study. Luna is cheap and leaky; Sol is dearer and still misses lexical oracles.

- Substring oracles are brittle. Sol C "use SQLite, not Redis" failed a phrase check with residual recall 3/3. We kept the oracle.
- Experimental cells are not in live_cases(). Factory-flip and last-wins receipts keep claim_ready false by construction (suite_incomplete). That is intentional.
- Layer 0 arm A is a scripted omission control. Do not cite it as "summarizers fail."
- The filler that taught the selector ("docs only", "pad pad pad") is lab-shaped. We tightened the filler rule after it dropped a real plan that used those words. Adversarial English remains open.
- Topic last-wins is noun overlap, not entailment. Two unrelated sentences that share enough content nouns can evict the earlier one. We have not bounded that failure.
- Vault is session-scoped BM25. It will not answer "what did we decide?" from tokens that are not in the story or the plan chunk. That is the point of the story.
- Costs are provider usage.cost sums on these receipts, not a general pricing model.
- This paper packages work that already shipped in a product. It is a public lab notebook, not an independent replication by a third party.

9 Reproduction

Hermetic battery (no API keys), from a Marionette checkout that still exposes the product seams, with this lab on PYTHONPATH:

```
python -m catalog_residual.bench
python -m catalog_residual.live          # dry-run
python -m catalog_residual.live --live \
  --driver openrouter:openai/gpt-5.6-luna \
  --suite all --repeats 3
```

Live arms are Table 1. –suite is core (default), holdout, or all. –case overrides and is how experimental cells run. Published aggregates are data/published_tables.json. Do not add a winner field. Do not put experimental cells into live_cases() to force claim_ready.

Product tests for residual modes, vault unit behavior, and Settings stay in Marionette. This repo owns the protocol and the paper.

10 Conclusion

We measured compaction residuals instead of arguing about them. Extractive handles plus a last-wins story are a credible factory default for a coding harness that already has a vault and an archive. A paid paragraph is still the better residual for some asks and is still one Settings click away. Hybrid is the ranking winner on the claim-grade suite and the honest opt-in when someone wants both channels in one residual.

The work is small, local, and reviewable. If the oracles are wrong, say so. If last-wins is too crude, patch it. If another harness replicates the factory-flip gate with a different model pair, that is the contribution we actually want.

Acknowledgments

Receipts were run against Marionette's own compact path, not a reimplementation. Puppetmaster is the orchestration kernel under that harness; this paper does not evaluate it. Errors are the author's.

References

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[3] S. Robertson and H. Zaragoza. The Probabilistic Relevance Framework: BM25 and Beyond. Foundations and Trends in Information Retrieval, 3(4):333–389, 2009.

[4] C. Palmer. Marionette v0.9.245. 2026. https://github.com/professorpalmer/marionette.

A Cost ledger (provider sums)

Table 6: Provider usage.cost totals from the published aggregates. These are the receipts, not estimates.

Receipt	n	USD
suite-all Sol (hybrid without stems)	120	1.006
suite-all Luna	120	0.098
hybrid+stems Luna slice	36	0.034
hybrid+stems Sol slice	36	0.341
claim-grade Luna	132	0.109
claim-grade Sol	132	1.125
claim-grade merge	264	1.234
factory-flip Luna	96	0.082
factory-flip Sol	96	0.830
summarizer last-wins Luna	12	0.011
summarizer last-wins Sol	12	0.094

B Gate predicates (informal)

Let S be the set of ok rows in a receipt. Write $r_1(x)$, $e(x)$, $p(x)$, $s(x)$ for round-1 residual recall, end-task success, peek calls, and peek stale. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be the compact arms and NC the honesty holdouts.

$$\text{honesty\_fail} = \exists\, x \in \mathrm{NC} :\; x \text{ invents the absent token} \tag{1}$$

$$\text{stale\_tax\_fail} = \Big(\sum_{x \in \mathcal{C}} p(x) > 0\Big) \wedge \frac{\sum s(x)}{\sum p(x)} > 0.25 \tag{2}$$

$$\text{saturation\_fail} = \big(\bar{e}(\mathcal{A}) \geq 0.95 \wedge \bar{e}(\mathcal{B}) \geq 0.95\big) \vee \big(\bar{r}_1(\mathcal{A}, \mathcal{B}, \mathcal{C}) \geq 0.95\big) \tag{3}$$

claim_ready additionally requires two models, a complete live_cases() factorial, residual text and final answer on every ok row, and no winner key. Experimental receipts fail the factorial on purpose.